Questions

11 Is a sensor an example of an input device or an output device?

12 Can you name **five** different types of sensor?

13 Can you think of one application for each of the following sensors?

- a Light
- b Magnetic field
- c Temperature

14 Can you think of **two** ways that the use of sensors is of benefit to humans?

15 How could an accelerometer be used to protect the hard drive on a computer?



Figure 3.8: Art galleries use temperature and humidity sensors

3.7 Data storage

You need to store data in a computer system on a constant basis. This might be temporarily, whilst it is being processed, or on a more permanent basis for future use. For these purposes, there are two main categories of storage that exist in a computer, they are **primary storage** and **secondary storage**.

Primary storage is a category of storage that is directly accessed by the CPU. This means that the CPU has a direct connection to it and that data is sent between the CPU and the primary storage using this connection. There are two main types of primary storage, random access memory (RAM) and **read only memory (ROM)**. There are several characteristics of each type of primary storage, these include those shown in Table 3.2:

Characteristics of RAM	Characteristics of ROM
It is volatile storage. This mean that it requires power to store data. If the power is turned off, the contents of RAM are lost. Volatile storage is sometimes referred to as temporary storage.	It is non-volatile. This means that if power is removed, the contents of the ROM are retained. Non-volatile storage is sometimes referred to as permanent storage.
It stores data and programs that are currently in use by the computer.	It stores programs that initially boot the computer, such as the bootstrap and the BIOS.
Its contents are constantly being changed as data is processed by the CPU.	The contents of ROM are fixed and do not normally change. It is possible to reprogram it, but this is a complex process and only normally attempted in circumstance when a computer system is failing to boot correctly.
It is possible to increase the amount of RAM a computer has by adding further RAM components	It is not easy to alter the ROM available in a computer system.

Table 3.2: Characteristics of primary storage

KEY WORDS

primary storage: a type of storage that is directly accessed by the CPU.

secondary storage: a type of storage that is designed to store data permanently, until it is deleted by the user.

ROM (read only memory): a type of primary storage that stores the start up instruction for the computer.

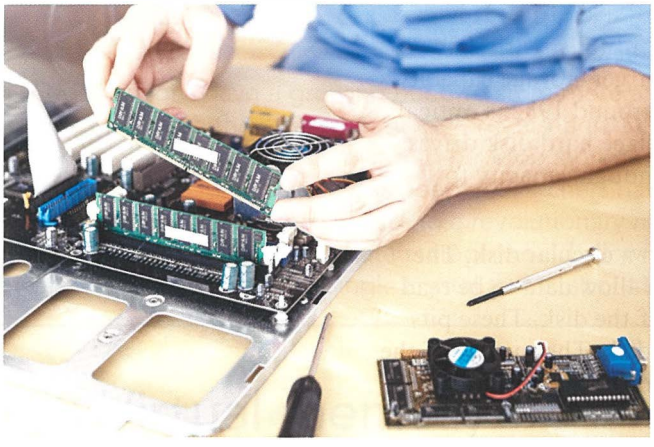


Figure 3.9: Technician changing a computer's RAM



Figure 3.10: The inside of a hard disk drive

ACTIVITY 3.6

Did you know that there are also different types of RAM and ROM? You do not need to know what these are for the exam, but it is interesting to find out what they are. Use the internet to research the different types of RAM and ROM and when they are used.

Secondary storage is a category of storage that is not directly accessed by the CPU. This means that data and instructions need to be sent from secondary storage into primary storage in order for it to be processed. Secondary storage is also a type of non-volatile storage. Can you remember what this means? There are three main types of secondary storage, these are **magnetic**, **optical** and **solid-state storage**.

Magnetic storage relies on the properties of magnetism to store data. A modern magnetic storage device normally has circular plates called platters that spin to allow data to be read and written. The platters are separated into tracks and sectors. It has a read/write head that moves across the platters to read or write the data. It uses an electromagnet to magnetise dots on each platter that store the data. If a dot is magnetised it stores the binary value 1. If a dot is demagnetised it stores the binary value 0. Each platter has billions of these magnetised dots. An example of this type device is called a hard disk drive.

ACTIVITY 3.7

You have just learnt that a hard disk drive is a type of magnetic storage and how it operates. There are older types of magnetic storage called magnetic tape and floppy disks. What can you find out about them?

Can you also find out what the largest storage capacity is for a hard disk drive?

KEY WORDS

magnetic storage: a type of secondary storage that uses the properties of magnetism to store data.

optical storage: a type of secondary storage that uses lasers to store data.

solid-state storage: a type of secondary storage that uses transistors to store data.

REFLECTION

After completing Activity 3.7:

What kind of search data did you use to find out the data for the activity? Do you think you could have improved your search data?

In 1956, a technology company called IBM sold the first hard disk drive. It had a storage capacity of just 5 MiB of data and cost $50 000! Thankfully, hard disk drives have become much cheaper to buy and have a much larger storage capacity now. Another type of storage that became available to buy around 1982 is optical storage. People originally thought optical storage would replace hard disk drives, but soon realised that they could not be made with a high enough capacity to do this.

Optical storage uses lasers to read and write data to and from a circular disk. The disk spins and a read/write heads moves across the surface to allow data to be read or written. To write data, a laser burns pits into the surface of the disk. These pits are organised in a spiral track that starts at the centre of the disk. The surface of the disk between each pit is called a land. A laser is also used to read the pits and lands to obtain the data that they store. Examples of optical media are a CD, DVD and a Blu-ray. Two of those examples are read using a red laser and one is read using a blue laser, can you guess which is which?

ACTIVITY 3.8

Draw a diagram to represent the operation of optical storage.

Although optical storage is still used today, it is mostly used to archive data or as a cheap way to physically distribute files to users that require them. They didn't replace hard drives in the way that it was first thought they would. As portable devices developed, people began to realise there was a disadvantage with hard disk drives. This disadvantage came from the fact that they have moving parts. This made them vulnerable to wear and tear and also damage if they were dropped. As a result, a new type of storage was developed called solid-state storage.

Solid state storage doesn't have any moving parts. Instead it is created using **semiconductor chips**. Solid-state storage is sometimes known as flash memory. This is because of the ways that the data is read and written to the chips, it is said that it is 'flashed' onto them, but how does that happen? Solid state storage contains **cells** and **transistors** that are laid out in a grid. The transistors can be wired together in two different ways. In a NOR flash structure, the cells are wired together in a parallel structure. In a NAND flash structure, the cells are wired together in series. The intersection of the rows and columns of transistors are the cells. One of the transistors is called a **control gate** and one of the transistors is called a **floating gate**. The control gate sits on top of the floating gate. When a solid state storage device is first created, all the transistors are set to the binary value 1. The electric current reaches each control gate and then flows through to the floating gate to be stored. When data is stored in a transistor, it is converted to the binary value 0. A solid state drive is one example of a solid state storage device. You probably use another example on a regular basis too, it's called a USB flash memory drive (also known as USB stick, memory stick or a thumb drive).

ACTIVITY 3.9

You have learnt that a solid state drive and a USB flash memory drive are example of solid state storage. Can you think of another example?



Figure 3.11: CDs are a type of optical media

KEY WORD

semiconductor chips: an electronic circuit, where transistors are set on a semiconductor material, such as silicon.

COMPUTER SCIENCE IN CONTEXT

Organisations have a growing need to store more and more data. This data is often vital to the operation of the organisation. Data storage development companies are constantly trying to develop storage devices that have larger storage capacities. Due to this need, researchers have started to look at other methods of data storage including a revolutionary method called holographic data storage. This type of data storage would allow approximately a terabyte of data to be stored in a single crystal that is about the size of a grain of sugar.



Figure 3.12: A USB flash memory drive

KEY WORDS

cell: the intersection of the rows and columns of transistors in solid-state storage.

transistor: a device that acts as a switch or gate for electronic signals.

control gate: a component that controls the flow of electric current.

floating gate: a component that can store electrical charge.

virtual memory: a type of memory that can be used as an extension to the RAM.

page: a unit of data.

3.8 Virtual memory

So far, all the storage that you have learnt about has been physical storage. This means that it is an actual component storage unit that you can see and touch. There is another type of storage that can be created in a computer and this is called **virtual memory**. It is called virtual memory, as unlike physical storage, it isn't a component that you can see or touch.

You may have noticed that this chapter includes the term memory and storage. You may find these used interchangeably in lots of different resources. There is a slight difference between the two, but even the defined difference is a little contradictory.

Technically, memory is used as a term for short term temporary storage of data, and storage is used as a term for the permanent storage of data. For example, RAM stores data temporarily and is a type of memory, random access memory. A hard disk drive is a type of storage as it stores data permanently, until the user deletes it. This seems a simple distinction, but ROM is also a type of memory, read only memory, but it stores data permanently too. Therefore, the difference is not completely clear, but it is mostly considered that memory is temporary and storage is permanent. However, you don't need to become too worried about picking the correct term!

Virtual memory is also a type of memory that is created for temporary use. It is created as an extension to RAM. Sometimes, a computer is given a large number of tasks to complete at a given time. This means that it will need a lot of data to do this. All this data will need to be sent to the RAM to be ready to be processed. A computer system only has a set amount of RAM, so what happens when this RAM becomes full? It needs some extra help! This is where virtual memory can help.

Virtual memory is created in a computer system with a hard drive component. A section of that hard drive is partitioned off and is effectively an extra bit for RAM instead. When the RAM gets full, the computer system can transfer some data to the virtual memory to be held there until it is needed. It will choose to send data that isn't as immediately required as other data. The data that is transferred is collected in a structure called a **page**. These pages of data are transferred to the virtual memory to make sure there is enough room in the RAM for the immediate data that is being processed. Once the pages of data are required again, they can be sent back to the RAM to be processed. The diagram in Figure 3.13 shows the process:

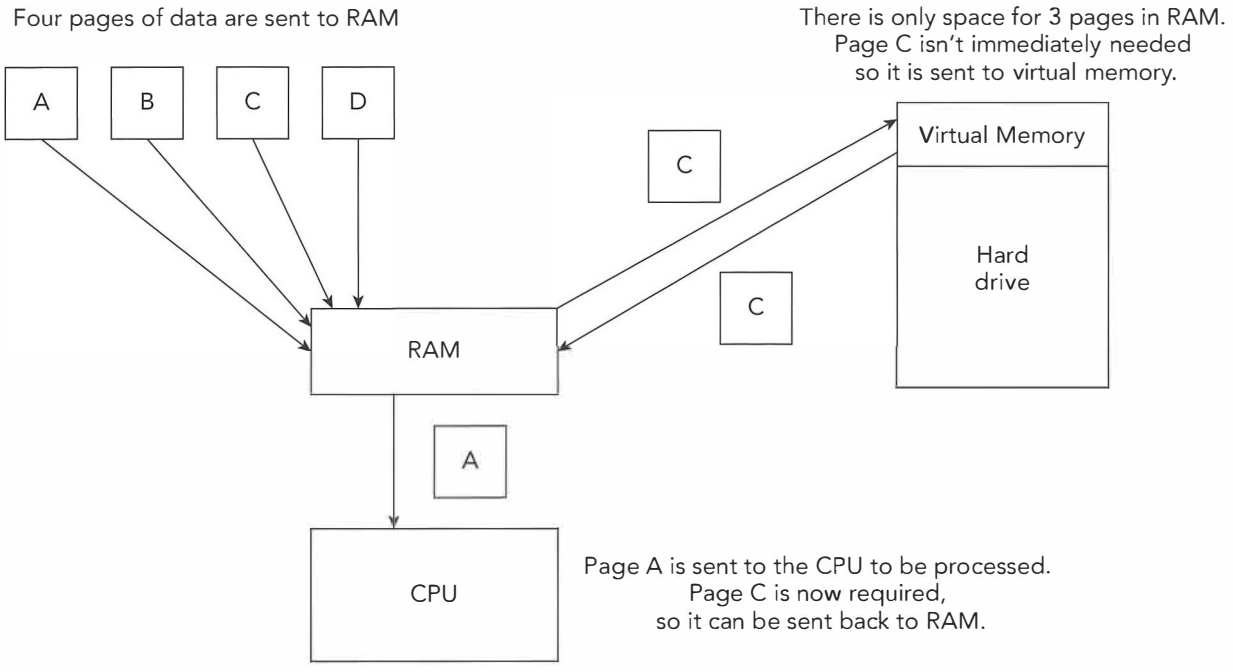


Figure 3.13: Pages of data are transferred to the virtual memory so that RAM doesn't become full

If the RAM became full and the computer did not have the ability to transfer pages of data to the virtual memory, this could cause it to crash, as it may not be able to complete the task it is currently working on.

3.9 Cloud storage

Despite a common misconception that many people have, **cloud storage** does not store our data somewhere up in the clouds. So how did it get that name? It is more in relation to the very large collections of data that are stored being like clouds of data, think clouds of millions of raindrops!

As technology has developed, and continues to do so, people have found that they want and need to store larger amounts of data, and data storage is quite expensive to buy, keep and maintain.

You might be able to make sure that your 2 Terabyte hard drive at home is safe in your house and is checked regularly for security issues, but it's likely you only have a relatively small amount of data to store. Some multinational businesses have huge amounts of data to store. This can require large rooms that are needed to house expensive hardware, such as computer **servers**, to store all the data that the company requires. This could be the details of millions of their customers and all the online transactions they have made to buy products. They would also need staff to maintain this equipment and keep that data safe. These kinds of costs could amount to many thousands of pounds a year for a business. Therefore, cloud storage offers them an alternative.

KEY WORDS

cloud storage: data storage that is owned by a third party and accessed by the user, using the internet.

server: a component that acts as a central resource for the storage of data and applications or for providing a service.

Cloud storage companies created a service in which they set up very large buildings that house and maintain thousands of computer servers. They spend 24 hours a day making sure the servers are maintained and secure. A company can then pay a fee to the cloud storage company for their service, to be allowed to store their data on their servers and pass the responsibility to the company to keep it secure until it is required. The company then accesses this data, when it is required, using the internet.

Cloud storage is a collection of third party hardware (hardware that is owned by another person or organisation), such as computer servers, that individuals and companies can use to store data, and for a range of other services. When data is stored within the buildings and hardware owned by the individual or business, this is known as storing the data locally. When data is stored in hardware owned by a third party and accessed using the internet, this is known as storing data remotely.



Figure 3.14: Data can be stored in cloud storage

Cloud storage began as a simple way to offload the responsibility of storing data. Since it was first used, it has developed into use for other services. For example, a business can now use cloud storage in connection to its website. The cloud storage company can offer a service whereby when demand is high for a website, such as they have a big sale, then they can use cloud storage to extend the capabilities of the web servers to be able to process the large amounts of data and requests that are being sent to the website. This means that the company can use hardware they do not own, on occasions when the demand for their business is high. This means they do not need to own all the hardware used, and have the extra hardware not being used, and therefore wasting money for them, when their business demand is lower. Helpful for business costs!

COMPUTER SCIENCE IN CONTEXT

Individuals are making increasing use of cloud storage. This allows portable devices to remain fairly small, as they do not need to store very large amounts of data locally within the portable device, e.g. their mobile phone. Companies such as Microsoft also offer access to software using cloud storage now, e.g. Office 365. This means that people do not need to download large software applications onto their portable devices.

Some people do have concerns about cloud storage. They think it is a big risk to give someone else the responsibility for looking after their data. Table 3.3 gives comparisons of some advantages and disadvantages of storing data in the cloud and storing it locally.

Storing data in the cloud	Storing data locally
You do not have to pay for the hardware to store the data.	You will need to purchase the hardware to store your data. These costs can be high if you have a lot of data to store.
You do not have the responsibility for the security of the hardware that stores the data. However, you are relying on a third party making sure that your data is safe. If they do not do this correctly, they may put your data at risk.	You will need to make sure that your data storage is well maintained and that it is regularly checked and updated for security issues.
You can access the data and resources using different devices from anywhere that you have an internet connection. However, if your connection fails or is not available, you will not be able to access your data. If your connection is slow, this can also mean that accessing your data is slow.	You do not need an internet connection to access your data, therefore it is available at all times that you have local access to that data.
You can increase and decrease the amount of storage space you need very easily, when you need to.	You might have redundant hardware, that you have paid for, that isn't used as the amount of data that you need to store has decreased.

Table 3.3: Comparisons of storing data in the cloud and locally

ACTIVITY 3.10

Do you store any data in cloud storage? If so, do you ever worry about doing this? Discuss with a partner about the advantages and disadvantages of storing data in the cloud, compared to local storage. Reach a conclusion about whether you think the risk is worth it.

As part of your discussion research at least one more advantage and one more disadvantage and make note of these. You can then share your thoughts from your discussion with the rest of your class and see who thinks cloud or local storage is better, and why.

3.10 Network hardware

At the beginning of this chapter, you learnt about the hardware required within a computer and local to a computer that enable the temporary and permanent storage of data. You have now learnt that data can be stored remotely, in a cloud storage system, but that an internet connection is required to access the data stored in the cloud. The hardware required for this connection creates a **network**. You will now learn what hardware is required to create a connection to other devices and the internet.

When two or more devices are connected, this is called a network. Some basic hardware devices are required to create a network, these are a **network interface card (NIC)**, a **router** and you may need some **transmission media**.

KEY WORDS

network: computers and devices that are joined together using cables or wireless technology.

network interface card (NIC): the component in a computer that is used to connect it to a network.

router: a network component that examines a data packet to obtain its destination address and then forwards the packet to this address.

transmission media: the cables that are used to transfer data in a network.